

YOLOv7 Object Detection Report

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1. Introduction

YOLOv7 is an object detection model introduced in 2022. It performs object detection and classification in a single forward pass, making it fast and suitable for real-time applications. In this task, YOLOv7 was used to detect different classes in the wheat dataset.

2. Environment Setup

The model was set up using a virtual environment. Required libraries such as PyTorch and other dependencies were installed. The YOLOv7 repository was cloned and configured for training.

3. Dataset Preparation

The dataset was organized in the required YOLO format:

```
dataset/
├── images/
│   ├── train
│   ├── val
│   └── test
└── labels/
    ├── train
    ├── val
    └── test
```

A custom wheat.yaml file was created with:

- **3 classes:** disease, pest, wheat
- Paths for train, validation, and test data

The dataset was manually split into training, validation, and testing sets.

4. Training Process

The model was trained using the following command:

```
python train.py --data data/wheat.yaml --weights yolov7.pt --epochs 50
```

Two types of weights were used:

- **Provided weights (given in assignment)**
- **Standard YOLOv7 weights (downloaded from official source)**

Comparison of Weights (Provided vs Standard YOLOv7)

1. Training Setup (Same for Both Experiments)

To ensure a fair comparison, both models were trained using the same hyperparameters:

- **Epochs:** 50
- **Batch Size:** 8
- **Image Size:** 640
- **Optimizer:** SGD
- **Learning Rate:** 0.01

2. Results Using Standard YOLOv7 Weights (Downloaded)

- **Precision (P):** 0.413
- **Recall (R):** 0.326
- **mAP@0.5:** 0.302
- **mAP@0.5:0.95:** 0.0955

3. Results Using Provided Weights (Assignment)

- **Precision (P):** 0.692
- **Recall (R):** 0.252
- **mAP@0.5:** 0.251
- **mAP@0.5:0.95:** 0.0801

Metric	Standard YOLOv7	Provided Weights
Precision	0.413	0.692 (higher)
Recall	0.326 (better)	0.252
mAP@0.5	0.302 (better)	0.251
mAP@0.5:0.95	0.0955 (better)	0.0801
Stability	Balanced	Unstable (class 1 failed)

Performance Analysis

Although the provided weights achieved a higher precision value, their overall performance was lower. This is because the recall and mAP values are more important indicators of detection quality.

The provided weights showed very poor recall, especially for one of the classes where recall was zero. This means the model failed to detect instances of that class entirely. As a result, despite high precision, the model is not reliable.

On the other hand, the standard YOLOv7 weights produced more balanced results across all classes, with better recall and higher mAP scores.

Reason Why Standard YOLOv7 Weights Performed Better

The standard YOLOv7 weights performed better due to the following reasons:

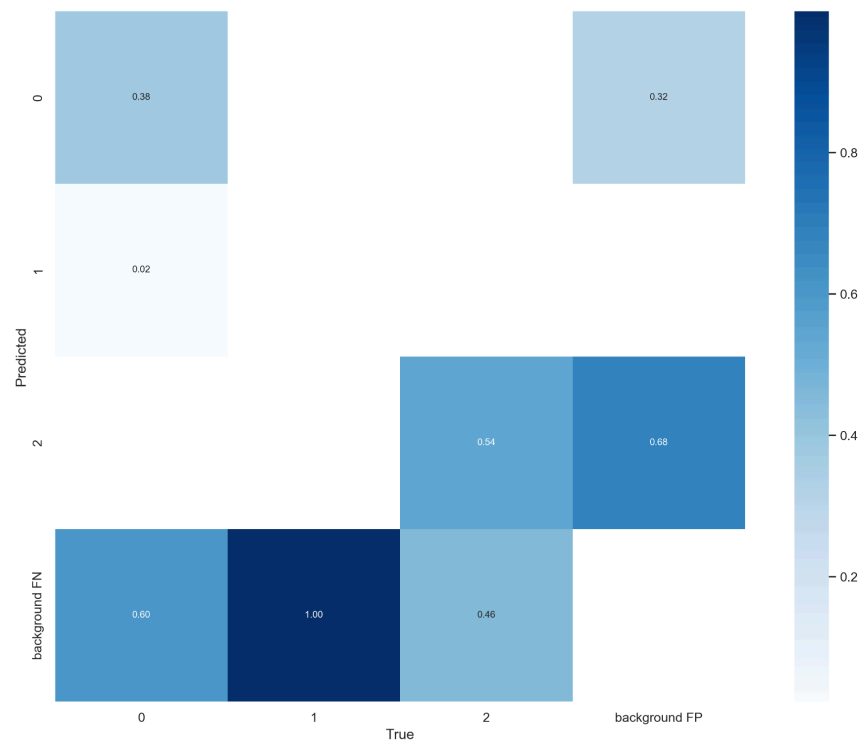
1. Pre-training on a large dataset
The standard YOLOv7 model is trained on the COCO dataset, which contains a wide variety of objects. This allows the model to learn strong and general features.
2. Better feature extraction
Since the model is already trained on diverse data, it can better recognize patterns such as shapes and textures, making fine-tuning more effective.
3. Higher recall
The standard weights were able to detect more objects in the images, resulting in higher recall values.
4. Better generalization
The provided weights likely lacked diversity in training, causing poor generalization and failure to detect certain classes.

Conclusion

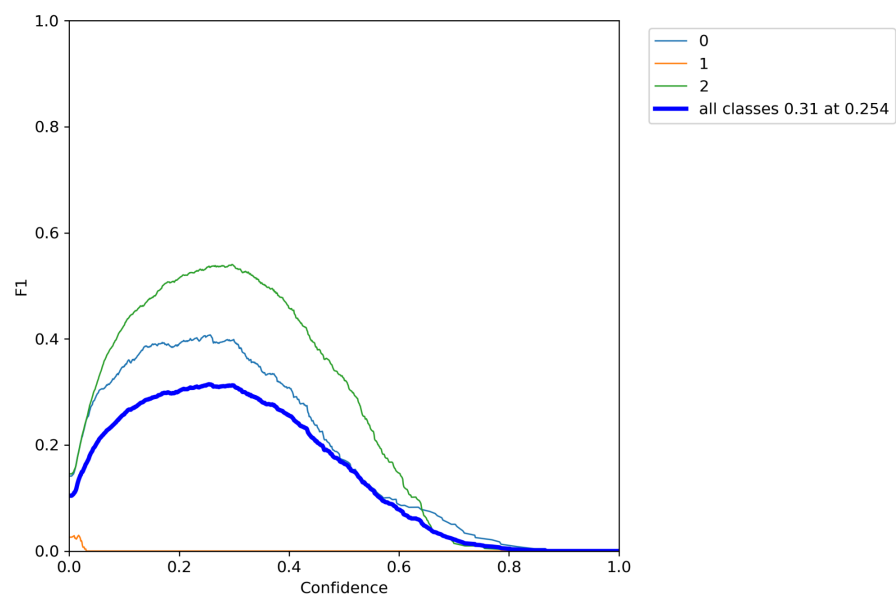
Although the provided weights achieved higher precision, the standard YOLOv7 weights performed better overall due to higher recall and mAP values. Therefore, the standard YOLOv7 weights are more suitable for this task as they provide more reliable and consistent object detection performance.

Plots

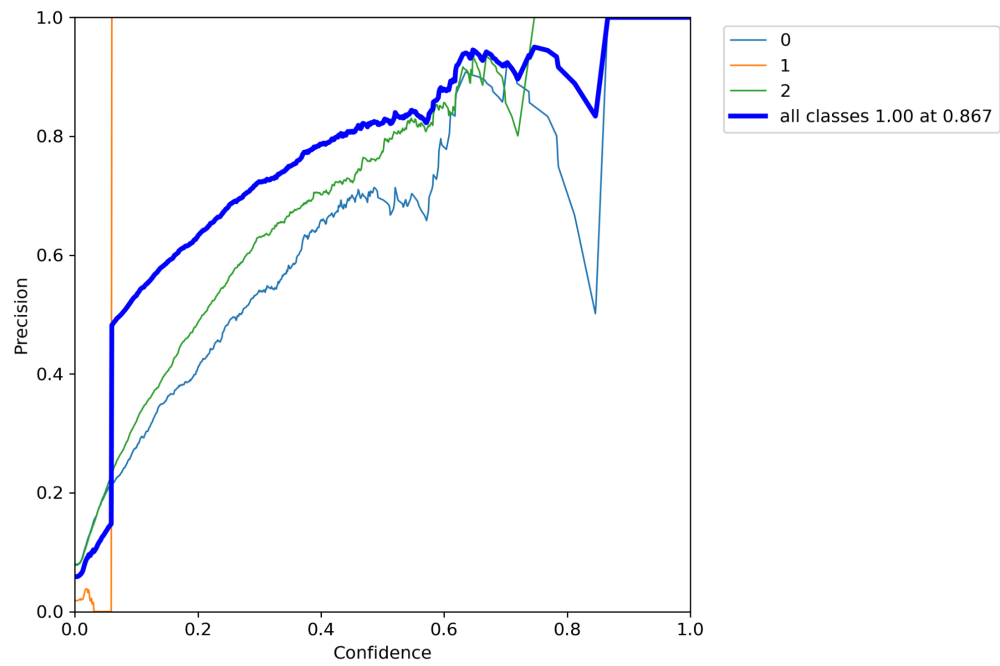
Confusion Matrix



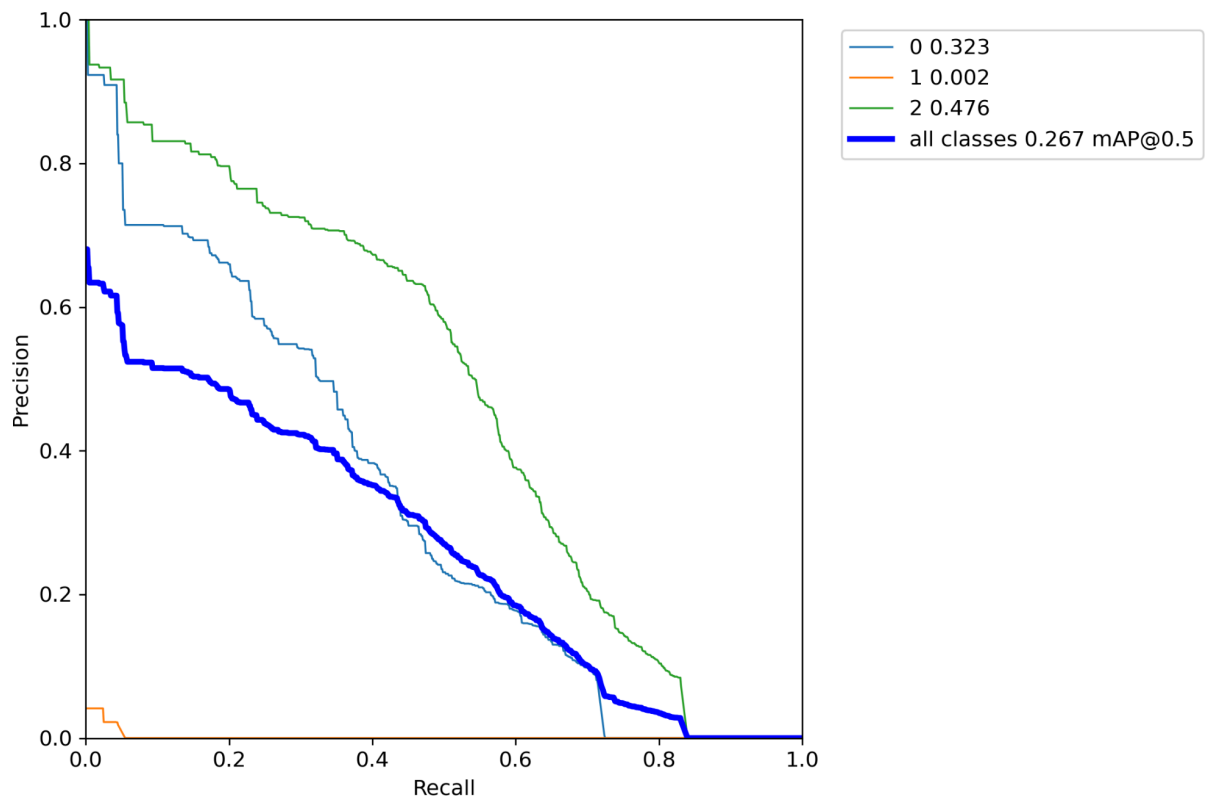
F1 Curve



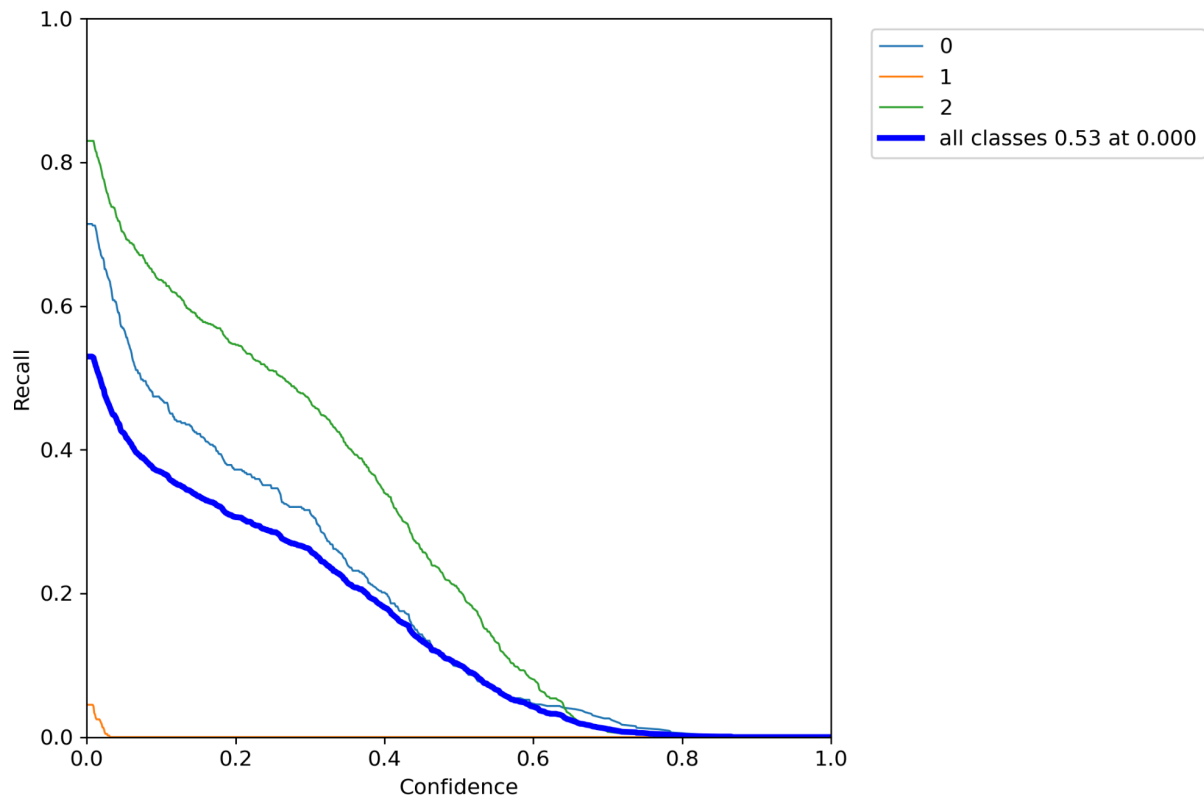
P Curve



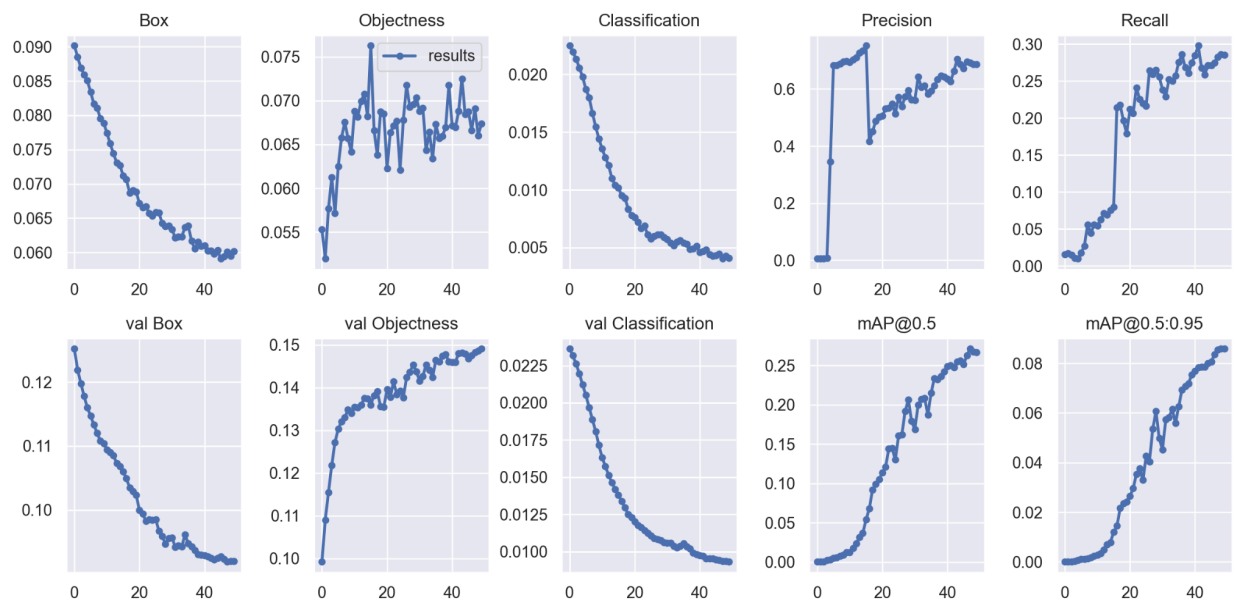
PR Curve



R Curve



Result



RGB images with ground truth detections and fine tuned detections side by side.

